4.10.12

EE25BTECH11011 - Navya

Question:
The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

Solution:
The equations of the given lines are

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \mathbf{x} = 10 \tag{1}$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \mathbf{x} = -5 \tag{2}$$

Writing (1) and (2) in augmented matrix form,

$$\begin{pmatrix} 1 & 2 & 10 \\ 2 & 1 & -5 \end{pmatrix} \tag{3}$$

Performing row operations,

$$R_2 \to R_2 - 2R_1 : \quad \begin{pmatrix} 1 & 2 & 10 \\ 0 & -3 & -25 \end{pmatrix} \tag{4}$$

$$R_2 \to -\frac{1}{3}R_2 : \quad \begin{pmatrix} 1 & 2 & 10 \\ 0 & 1 & \frac{25}{3} \end{pmatrix} \tag{5}$$

$$R_1 \to R_1 - 2R_2 : \quad \begin{pmatrix} 1 & 0 & -\frac{20}{3} \\ 0 & 1 & \frac{25}{3} \end{pmatrix} \tag{6}$$

Thus, the intersection point is

$$\mathbf{x} = \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} \tag{7}$$

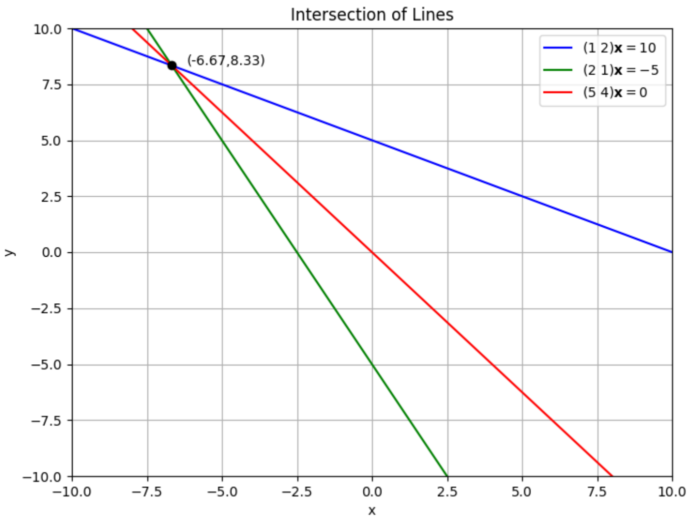
Now, check whether it lies on the line

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \mathbf{x} = 0 \tag{8}$$

Substituting,

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} = \frac{1}{3}(-100 + 100) = 0 \tag{9}$$

Hence, The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.


Intersection of Lines
(-6.67,8.33)
(1 2)x = 10
(2 1)x = −5
(5 4)x = 0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x
y